By defn. 1, $(m+p)$ is an even integer. $\boxed{\therefore r \text{ is True}}$ □

6) Use a direct proof to show that the product of two odd numbers is odd.

Soln: p: x is an arbitrary odd no. q: y is an arbitrary odd no.

r: xy is an odd no.

We need to give a direct proof that $(p \wedge q) \to r$ is True.

Let, $(p \wedge q)$ be True. By defn. of AND, p, q are True each.

$\therefore x$ and y are both arbitrary odd no.s

By defn 1, there exists two ~~[illegible]~~ integers K_1 and K_2 such that

$x = 2K_1+1$ and $y = 2K_2+1$

$\therefore xy = (2K_1+1)(2K_2+1) = 4K_1K_2 + 2K_1 + 2K_2 + 1 = 2(2K_1K_2 + K_1 + K_2) + 1$

$\therefore K_1, K_2$ are integers $2K_1K_2 + K_1 + K_2$ is also an integer.

By defn. of odd integer in defn 1, xy is odd. $\boxed{\therefore r \text{ is True}}$ □

7) Use a direct proof to show that every odd integer is the difference of two squares.

Soln: p: x is an odd integer q: There exists two integers a and b such that $a-b=x$, and $a=m^2, b=n^2$ for some integer m, n

We shall give a direct proof that $p \to q$ is True.

Let p be True. $\therefore x$ is an arbitrary odd integer. By defn 1, there exists an integer K s.t. $x = 2K+1 \Rightarrow x = (K^2+2K+1) - K^2 = (K+1)^2 - K^2$

$\therefore K$ is an integer, $K+1$ is also an integer. K^2 and $(K+1)^2$ are also integers.

$\boxed{\therefore x \text{ is the difference of two squares and } q \text{ is True}}$ □

8) Prove that if n is a perfect square, then $n+2$ is not a perfect square.

Soln: p: n is a perfect square q: $n+2$ is not a perfect square

We shall prove $p \to q$ is True by contradiction